

Place-Based Credit inside Functional Regional Economies: Spatial Incidence, Reallocation, and Urban Hierarchy

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Abstract. Because workers, firms, suppliers, and credit relationships cross municipal borders, a local policy gain can create regional activity, displace activity from nearby places, or change which tier of the regional hierarchy captures formal employment and establishments. This paper studies that spatial-incidence problem using Brazil’s Northeast Constitutional Financing Fund (FNE), a large place-based credit program that changes the cost and availability of productive capital across eligible municipalities. Predicted exposure combines predetermined sectoral structure with external sector-year movements in subsidized credit, and the baseline design compares municipalities within the same Immediate Region and year. Exposed municipalities expand formal employment, establishments, and payroll; the rest of the region does not contract; regional formal aggregates rise; and exposed municipalities gain shares of regional formal activity. Effects are concentrated outside regional cores and strengthen among less central or smaller non-core municipalities. Because regional value added and value-added intensity do not rise, the evidence shows that place-based credit expands and decentralizes formal scale more clearly than it generates regional value-added upgrading.

Keywords: regional science, spatial incidence, place-based credit, functional regional economies, regional hierarchy, development finance, shift-share exposure

JEL Classification: G21; R11; R12; R58

1. Introduction

Regional economies connect jurisdictions through workers, firms, suppliers, consumers, credit relationships, and public services, so a gain measured in one municipality can have several regional meanings. It may reflect new activity in the functional regional economy, displacement from nearby municipalities, diffusion through local supply and labor-market linkages, or a shift in the regional hierarchy. For that reason, treated-place estimates leave a central regional-science question unresolved: how does a functional regional economy absorb a local policy shock?

Credit policy makes this spatial-incidence problem especially sharp because subsidized productive credit changes the cost and availability of capital. A capital-cost shock can expand constrained firms and producers, while also altering the relative attractiveness of municipalities within the same

regional system. When financial frictions bind more strongly outside regional cores, credit may allow less central municipalities to host formal activity that would otherwise remain concentrated in dominant centers. When firms and workers are highly mobile within the region, local gains may instead reflect intraregional reallocation. Distinguishing regional creation from spatial reshuffling is therefore essential for interpreting place-based credit.

Taken together, the regional-science literature provides the building blocks for this question. Work on regional systems and urban hierarchy emphasizes that size alone is spatially incomplete because a place’s position in the system determines access to services, labor markets, and agglomeration benefits [Moretti, 2011, Monte et al., 2018, Redding and Rossi-Hansberg, 2017, Soto and Paredes, 2016]. Research on the geography of credit shows that access to finance is itself spatially uneven, shaped by urbanization, local banking markets, information, collateral, and regional institutions [Guiso et al., 2004, Cetorelli and Strahan, 2006, Carmignani et al., 2021, Rodríguez-Pose et al., 2021]. Evidence on place-based programs further shows that local incentives can generate gains at the treated place while leaving open who captures those gains and whether nearby places adjust [Busso et al., 2013, Kline and Moretti, 2014, Neumark and Simpson, 2015, Theodos et al., 2022, Rupasingha et al., 2024].

Brazil’s Northeast Constitutional Financing Fund (FNE) provides an empirical laboratory for this regional-incidence question. Operated by Banco do Nordeste, the program channels subsidized productive credit to municipalities in Brazil’s Northeast and adjacent areas covered by the regional development mandate [Brazil, 1989, Banco do Nordeste do Brasil, 2024a]. Its empirical value comes from a large, geographically targeted credit shock that varies across municipalities and sectors within a historically lagging macro-region. Municipalities are the exposure cells, and Immediate Regions—official functional regions designed to capture everyday regional economic interactions—define the incidence unit [Instituto Brasileiro de Geografia e Estatística, 2017].

To isolate the policy-induced component of local credit conditions, the analysis uses predicted exposure in place of realized municipal lending. I interact predetermined municipal sectoral shares with external sector-year movements in FNE credit, following the logic of shift-share exposure designs [Bartik, 1991, Autor et al., 2013, Adão et al., 2019,

Goldsmith-Pinkham et al., 2020, Borusyak et al., 2022]. Predicted exposure rises when a municipality's initial sectoral structure loads more heavily on sectors that later receive stronger FNE expansions outside the municipality's own state. Municipality fixed effects absorb fixed local characteristics, and Immediate-Region-by-year fixed effects compare municipalities inside the same functional regional economy at the same time. First-stage validation, alternative shocks, sector-cycle controls, leave-one-sector-out exercises, AKM and AKM0 inference, and Rotemberg-style diagnostics probe the design.

To turn local effects into regional incidence, the paper estimates a spatial-accounting framework with four margins: the exposed municipality, the rest of its Immediate Region, the entire region, and the municipality's share of regional activity. Predicted exposure raises formal employment, establishments, and payroll in exposed municipalities; the rest of the region does not contract; regional formal aggregates rise; and exposed municipalities gain regional shares. A one-standard-deviation increase in exposure implies about 10 percent higher municipal employment, roughly 297 additional formal jobs in the average municipality, and about 11 percent higher regional employment, roughly 3,700 jobs in the average Immediate Region. Taken together, these margins are inconsistent with a pure displacement story. Place-based credit expands regional formal scale while shifting regional weight toward more exposed municipalities.

Once the accounting separates regional creation from local footprint, regional hierarchy identifies where the new formal scale appears. Effects are small or absent in regional cores, while formal employment and establishment responses strengthen among non-core municipalities with lower initial positions in the regional employment distribution. A size gradient points in the same direction: initially smaller non-core municipalities experience the largest percentage gains and also gain regional shares. Value added disciplines the interpretation because municipal value-added estimates are positive in baseline specifications, but regional value added and regional value-added intensity do not rise. Place-based credit therefore creates and decentralizes formal scale more clearly than it generates regional value-added upgrading.

By separating spatial incidence from local footprint, the results contribute to regional science in three connected ways. First, it separates treated-place effects from spatial incidence inside the relevant functional regional economy, complementing work on local gains, efficiency, and spillovers in place-based policy [Busso et al., 2013, Kline and Moretti, 2014, Greenstone et al., 2010, Lu et al., 2019, Theodos et al., 2022, Rupasingha et al., 2024]. Second, it connects credit constraints to regional allocation by showing that subsidized capital should be evaluated not only by local employment or payroll responses, but also by where those responses occur within the regional system [Khawaja and Mian, 2008, Paravisini, 2008, Carmignani et al., 2021, Rodríguez-Pose et al., 2021, Van Doornik et al., 2024]. Third, the hierarchy gradient links place-based credit to the internal organization

of regional economies, where non-core municipalities may gain formal scale without a corresponding rise in regional productivity intensity [Duranton and Puga, 2004, Combes et al., 2012, Gaubert, 2018, Soto and Paredes, 2016, Duranton and Venables, 2018].

The remainder of the paper is organized as follows. Section 2 describes the institutional setting. Section 3 presents the data and measurement choices. Section 4 lays out the empirical strategy and spatial accounting framework. Section 5 reports the main results. Section 6 presents robustness checks and shift-share diagnostics. Section 7 concludes.

2. Institutional Setting: Credit Allocation across Functional Regional Economies

Created by Law 7,827/1989, the FNE is one of Brazil's constitutional regional development funds and channels long-term finance toward lagging areas in the Northeast and selected municipalities in neighboring states under the regional development mandate [Brazil, 1989]. Banco do Nordeste operates the fund by receiving loan applications, evaluating projects, and contracting credit across municipalities, sectors, borrower types, and credit lines. Although program rules evolve over time, the policy logic is stable: subsidized finance supports productive activity in places where private credit markets are less able to supply long-term capital at comparable terms [Banco do Nordeste do Brasil, 2024a,b].

The FNE works through a productive-credit channel that finances investment, modernization, working capital, and related production costs for firms, rural producers, cooperatives, associations, and microentrepreneurs. Agriculture, industry, commerce, and services all appear in the program, and that breadth matters for empirical design. A program financing rural producers and small businesses cannot be represented only by formal manufacturing employment, nor should its exposure be forced into detailed industrial classifications through discretionary concordances. Baseline exposure therefore follows broad FNE-compatible sectors, while alternative constructions are used as diagnostics.

Because eligibility is spatially defined, the fund is directly relevant for regional science. Municipalities inside the development area are organized around regional cores that concentrate services, transport access, markets, and administrative functions, so credit reaching one municipality can alter not only that municipality's scale but also its position relative to nearby municipalities. A loan-financed expansion in a peripheral municipality may create new formal jobs, support local suppliers, reduce dependence on the regional core, or formalize activity previously missing from administrative records. Distinguishing those margins is the empirical task of the paper.

Administrative detail creates useful variation because FNE operations are observed at the municipality-sector-year level and sectoral credit movements vary over time. Actual municipal lending remains endogenous, since credit demand, bank screening, local project quality, and contemporaneous

shocks can influence take-up. A predicted-exposure measure based on predetermined sectoral structure and external sector-year movements in FNE credit addresses that problem. Institutional variation supplies the shock, and within-region comparisons supply the spatial interpretation.

The main exposure measure excludes infrastructure credit to keep treatment close to private productive margins such as formal employment, establishments, payroll, and value added. This choice also avoids mixing credit to firms and producers with large public or quasi-public investment projects. Total FNE exposure, leave-one-municipality shocks, winsorized shocks, and leave-one-sector-out exposures are reported later, so the results do not depend on a single operational definition. Taken together, the setting is suited to asking whether place-based credit creates regional formal scale, reallocates activity within functional regions, or changes the regional hierarchy.

3. Data and Measurement

The measurement strategy is anchored in four administrative and official data sources. Municipal FNE operations come from Banco do Nordeste and provide annual credit by municipality and broad sector. Labor-market outcomes come from the *Relação Anual de Informações Sociais* (RAIS), the Brazilian employer-employee administrative records that cover formal establishments and jobs [Ministério do Trabalho e Emprego, 2024]. Municipal value added and GDP measures come from the Brazilian statistical office [Instituto Brasileiro de Geografia e Estatística, 2024]. These value-added measures are currently available through 2021, whereas labor-market outcomes are available through 2024. Spatial units come from the 2017 regional division of Brazil and from REGIC, the official study of urban influence and hierarchy [Instituto Brasileiro de Geografia e Estatística, 2017, 2020].

The estimation panel covers 1,898 municipalities observed from 2010 to 2024. Municipalities are nested in 166 Immediate Regions, which provide the baseline definition of functional local labor markets because they were designed to capture daily urban-regional articulation in access to goods, services, and central places. Regional aggregates are built by summing municipal outcomes within each Immediate Region-year, and regional hierarchy is measured using indicators that flag regional cores and poles.

Immediate Regions define the incidence unit as well as the main comparison frame. Municipalities are the exposure cells, but Immediate Regions are the functional systems in which local gains can appear as regional creation, contraction in the rest of the region, changes in municipal shares, or shifts in the regional hierarchy. This distinction matters because the paper's central object is neither the municipality alone nor the state-level regional economy; it is the way a functional regional system absorbs place-based credit exposure.

To match the spatial-accounting structure, the outcome set is organized around local, regional, share, intensity, and hierarchy margins. Local outcomes include log private employment, log private establishments, log real private payroll, and log

real private value added. Spatial-accounting outcomes add three margins to the local response: the log outcome in the rest of the Immediate Region excluding municipality i , the log outcome in the entire Immediate Region, and the municipality's share of the regional aggregate in percentage points. Intensity outcomes include employment per establishment, payroll per worker, value added per worker, and value added per establishment. Hierarchy gradients classify municipalities by their initial position in the regional employment distribution and by initial employment size. Predicted exposure is standardized to have mean zero and unit variance within the estimation sample, so coefficients are interpreted as effects of a one-standard-deviation increase in exposure.

Summary statistics in Table 1 describe the estimation sample before the regression analysis begins.

Two layers of variation are central to the empirical design and to the interpretation of regional incidence. Across municipalities, the panel contains large differences in formal labor-market size and in each municipality's share of regional activity. Across Immediate Regions, aggregate employment, establishments, payroll, value added, and internal concentration also vary substantially. Those layers allow the paper to ask separately whether exposure changes local outcomes, whether it changes the allocation of activity within the region, and whether the region expands in aggregate.

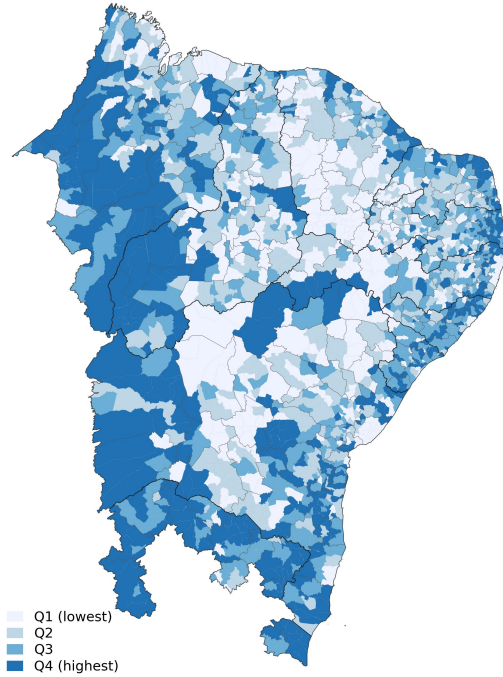
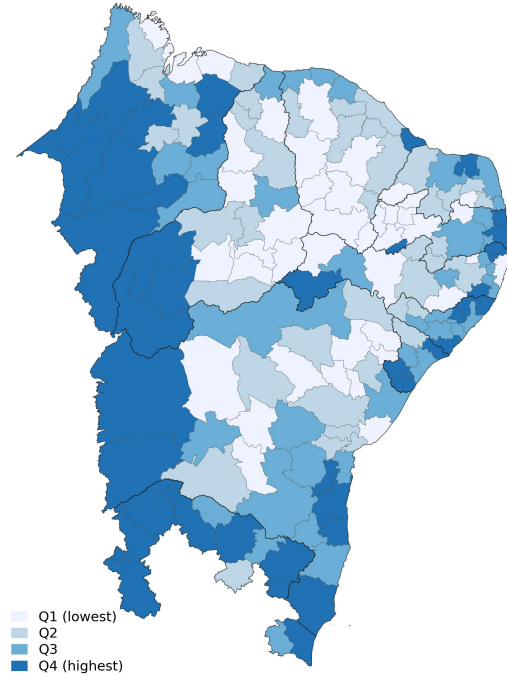
Geographic variation in Figure 1 provides the spatial counterpart to the descriptive table. Panel A maps average municipal exposure, while Panel B aggregates exposure to Immediate Regions. The two panels clarify why the empirical design uses both municipalities and functional regions: municipal variation identifies local and within-region incidence, whereas regional aggregation is used to ask whether formal activity expands in the aggregate.

Although the map is descriptive, it shows that exposure varies at the municipal scale and that Immediate Regions provide a meaningful spatial frame for aggregation and comparison. The empirical strategy below uses that structure to separate local footprint, rest-of-region responses, regional creation, and regional hierarchy.

Table 1: Descriptive statistics

Variable	Mean	SD	Min	Max	N
<i>Panel A. Municipality-year outcomes</i>					
Predicted exposure (municipal)	0.000	1.000	-2.021	6.369	28,470
Log private employment	5.641	1.962	0.000	13.478	28,470
Log private establishments	4.128	1.474	0.000	10.795	28,470
Log private payroll	13.163	2.120	0.000	21.454	28,470
Log private value added	18.390	1.399	15.177	25.048	22,772
Employment share within region (p.p.)	8.746	17.598	0.000	96.880	28,470
Establishment share within region (p.p.)	8.746	15.769	0.000	94.350	28,470
<i>Panel B. Region-year outcomes</i>					
Predicted exposure (region)	0.000	1.000	-2.670	5.269	2,490
Log regional employment	9.247	1.390	5.811	13.791	2,490
Log regional establishments	7.285	1.142	4.344	11.074	2,490
Log regional payroll	16.885	1.457	13.452	21.869	2,490
Log regional value added	21.435	1.179	18.742	25.588	1,992
Employment Herfindahl index	0.442	0.181	0.145	0.939	2,490
Pole employment share (p.p.)	59.370	18.413	11.603	96.880	2,490

Notes: The municipality-year panel covers 1,898 municipalities from 2010 to 2024. The region-year panel covers 166 Immediate Regions over the same period. Exposure measures are standardized within the estimation sample. Value-added outcomes use the available IBGE municipal accounts sample through 2021.

A Municipal exposure**B** Immediate-Region exposure**Figure 1:** Spatial distribution of predicted exposure to subsidized credit

Notes: Panel A maps the average value of predicted municipal exposure to FNE credit over 2010–2024. Panel B aggregates exposure to the Immediate-Region level. Darker shades indicate higher average exposure. Thin boundaries in Panel A denote municipalities, while thicker boundaries in both panels denote Immediate Regions and state borders.

4. Empirical Strategy and Spatial Accounting

Municipal exposure is built as a shift-share interaction between predetermined sectoral composition and external sector-year movements in FNE credit. Let s index broad program sectors, i municipalities, and t years. The exposure

measure is

$$Exposure_{it} = \sum_s share_{is0} \Delta FNE_{s,-uf(i),t}, \quad (1)$$

where $share_{is0}$ is the initial share of sector s in municipality i , while $\Delta FNE_{s,-uf(i),t}$ measures the sector-year movement in FNE credit computed outside the municipality's own state. This leave-state-out construction reduces mechanical

feedback from local and state-level credit allocation into the predicted exposure assigned to the municipality. Broad program-compatible sectors are used in the baseline so that treatment follows the institutional structure of the FNE and avoids a discretionary concordance with detailed industrial codes.

Exposure relevance is assessed through the validation equation

$$FNE_{it} = \alpha_i + \lambda_{rt} + \pi Exposure_{it} + u_{it}, \quad (2)$$

where FNE_{it} is observed municipal FNE credit or the municipality's share of regional FNE credit, α_i are municipality fixed effects, and λ_{rt} are Immediate-Region-by-year fixed effects. The validation equation links predicted exposure to observed municipal credit incidence, which is the relevant first-stage object for a design comparing municipalities within the same functional region.

Local effects are then estimated from

$$Y_{it} = \alpha_i + \lambda_{rt} + \beta Exposure_{it} + \varepsilon_{it}. \quad (3)$$

Spatial-incidence effects use the same specification after replacing the dependent variable with the municipality's share of the corresponding Immediate-Region aggregate, so β measures how the municipality's position within the regional distribution changes when predicted exposure rises.

The spatial-accounting exercise adds the rest of the region to the local and regional margins. For each municipality i in region r , define

$$Y_{irt}^{rest} = \log \left(1 + \sum_{j \in r, j \neq i} y_{jrt} \right). \quad (4)$$

Rest-of-region outcomes are estimated with municipality fixed effects and state-year fixed effects. Equation (4) serves as a spatial-accounting diagnostic because it asks whether exposed-municipality gains are accompanied by contraction, stability, or growth in the remaining part of the functional region. Regional-creation effects are estimated from

$$Y_{rt} = \alpha_r + \lambda_{ut} + \delta Exposure_{rt} + v_{rt}, \quad (5)$$

where Y_{rt} is a region-year aggregate, α_r are Immediate-Region fixed effects, and λ_{ut} are state-year fixed effects.

Regional hierarchy is studied by allowing the exposure effect to vary across groups defined by initial position in the regional system:

$$Y_{it} = \alpha_i + \lambda_{rt} + \sum_g \beta_g (Exposure_{it} \times G_i^g) + \eta_{it}, \quad (6)$$

where G_i^g denotes mutually exclusive groups such as regional cores, high-share non-core municipalities, middle-share non-core municipalities, and low-share non-core municipalities. A binary core/pole specification is retained as a benchmark in the appendix, but the gradient is the main regional-hierarchy object because it tests whether effects strengthen as municipalities move down the initial regional hierarchy.

Identification requires that, within the same Immediate Region and year, municipalities with different predetermined sectoral exposure would not have experienced differential changes aligned with external FNE sector-year movements in the absence of the program. The diagnostics target five threats to the design. External FNE movements might proxy for sectoral demand cycles; one broad sector or a small number of shock-cells might dominate the design; conventional clustered standard errors might understate uncertainty when municipalities share common shocks; regional shares might move mechanically with denominators; and large capitals or regional cores might account for the results. Municipal first-stage validation, sector-cycle controls, leave-one-sector-out estimates, AKM and AKM0 shift-share inference, Rotemberg-style diagnostics, alternative share denominators, alternative spatial fixed effects, and sample restrictions address those threats.

5. Results

The evidence is organized as a spatial decomposition because the same local coefficient has different regional meanings depending on what happens elsewhere in the functional region. Credit incidence first checks that predicted exposure maps into observed municipal lending. Spatial accounting then places four margins side by side: the exposed municipality, the rest of the Immediate Region, the entire Immediate Region, and the municipality's within-region share. Economic magnitudes translate the estimates into jobs, establishments, payroll, and relative share gains. Regional intensity estimates separate formal-scale creation from value-added upgrading. A hierarchy gradient finally asks where inside the regional system the new formal scale appears.

5.1. Exposure validation

Credit incidence is the bridge between the institutional policy and the empirical exposure measure. Realized municipal credit is not used directly in the outcome equations because loan demand, bank screening, local project quality, and contemporaneous municipal shocks can all affect actual lending. Predicted exposure instead isolates the component of the credit environment generated by predetermined sectoral structure and external sector-year movements in FNE credit. Table 2 verifies that this predicted component is connected to observed municipal credit.

Predicted exposure is positively related to observed non-infrastructure FNE credit and to total FNE credit. Share specifications are less precise, but they point in the same direction: more exposed municipalities capture a larger share of regional credit flows. This municipal validation is the relevant first-stage object because the paper studies municipal outcomes and municipal positions inside functional regions.

Table 2: Predicted exposure and observed municipal FNE credit

	(1) Log FNE non-infra	(2) Log FNE total	(3) FNE share non-infra	(4) FNE share total
Predicted exposure	0.077** (0.032)	0.068** (0.032)	0.295* (0.164)	0.214* (0.123)
Observations	28,470	28,470	28,470	28,470
Municipality FE	Yes	Yes	Yes	Yes
Immediate region $\times$ year FE	Yes	Yes	Yes	Yes
Clusters	166	166	166	166

Notes: Columns 1 and 2 use observed municipal FNE credit in logs. Columns 3 and 4 use the municipality's share of Immediate-Region FNE credit, in percentage points. Columns 1 and 3 use the baseline non-infrastructure exposure. Columns 2 and 4 use the corresponding total-FNE exposure. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.2. Spatial accounting inside functional regional economies

Spatial accounting provides the main empirical test because a local effect alone cannot distinguish regional creation from displacement, and a regional aggregate alone cannot identify which municipalities capture the new scale. Table 3 therefore reports the four objects that define spatial incidence: the exposed municipality, the rest of the Immediate Region, the entire Immediate Region, and the exposed municipality's share of the regional aggregate.

The main spatial pattern comes from formal labor-market outcomes, where all four margins point in the same direction. Employment, establishments, and payroll rise in exposed municipalities; regional employment, establishments, and payroll also rise; and exposed municipalities gain regional shares. At the same time, the remaining part of the Immediate Region does not contract. Employment and establishments in the rest of the region increase modestly, while payroll remains statistically flat. These estimates are inconsistent with a pure displacement story in which subsidized credit merely moves a fixed stock of formal activity from neighboring municipalities to exposed ones.

For regional science, this distinction is central because local jurisdictions are linked through commuting, supplier relationships, consumer markets, and regional services. Consequently, treated-place effects are not sufficient to interpret territorial policy [Monte et al., 2018, Suárez Serrato and Zidar, 2016, Fajgelbaum et al., 2019]. By measuring the exposed municipality, the rest of the functional region, the regional aggregate, and the within-region share in the same empirical framework, the table identifies the spatial incidence of place-based credit. The evidence points to formal-scale creation accompanied by a reallocation of regional weight toward more exposed municipalities.

Value added sets the boundary of that interpretation because the formal-scale result does not translate into regional upgrading. Exposed municipalities gain value added and value-added share in the baseline specification, and the rest-of-region value-added measure rises, but the regional aggregate does not increase. Hence the accounting exercise supports a strong claim for formal activity and a weaker claim for regional value-added upgrading. For interpretation, that con-

trast matters: place-based credit reorganizes the geography and scale of formal activity more clearly than it raises regional value-added intensity.

5.3. Economic magnitudes

Economic size matters for policy interpretation because a regional-incidence estimate should also indicate whether the changes are large enough to alter municipal positions inside regional systems. Coefficients in logs and shares are useful for inference, and Table 4 converts the main estimates into percent effects, jobs, establishments, payroll, and share changes using baseline 2010 levels.

For a one-standard-deviation increase in exposure, municipal employment rises by about 10 percent. At baseline means, this is roughly 297 additional formal jobs in the exposed municipality; at the median, it is about 16 jobs, a non-negligible change in small local labor markets. Regional employment rises by about 11 percent, or roughly 3,700 jobs in the average Immediate Region and 847 jobs in the median region. These magnitudes indicate that the formal-scale response is not only statistical precision in a large administrative panel.

Share effects carry the spatial interpretation because they show how exposure changes the relative position of municipalities within their functional regions. A 0.462 percentage-point gain in employment share equals roughly 32 percent of the median municipality's initial employment share. A similar logic applies to payroll share, where the gain equals about 36 percent of the median initial share. Place-based credit therefore changes the relative position of exposed municipalities inside their functional regions, not only their absolute level of formal employment.

5.4. Formal scale without value-added upgrading

Regional aggregates and intensity margins refine the accounting evidence by separating job creation from productivity upgrading. Place-based policy can plausibly affect one margin without affecting the other. Table 5 asks whether more exposed regions become larger and whether that additional scale is accompanied by higher payroll or value-added intensity.

Regional responses are concentrated in formal scale: employment, establishments, payroll, and employment per estab-

Table 3: Spatial accounting of predicted exposure

Outcome	Exposed municipality	Rest of Immediate Region	Entire Immediate Region	Within-region share
Employment	0.097*** (0.015)	0.013** (0.006)	0.105*** (0.016)	0.462*** (0.084)
Establishments	0.033*** (0.006)	0.008** (0.004)	0.042*** (0.009)	0.195*** (0.040)
Payroll	0.104*** (0.023)	0.012 (0.008)	0.097*** (0.020)	0.480*** (0.102)
Value added	0.054*** (0.017)	0.024** (0.010)	-0.006 (0.062)	0.394*** (0.146)

Notes: The exposed-municipality and within-region-share columns use municipality fixed effects and Immediate-Region-by-year fixed effects. The rest-of-region column uses the log of one plus the corresponding aggregate in the Immediate Region excluding municipality i , with municipality fixed effects and state-year fixed effects. The entire-region column uses Immediate-Region-year aggregates with Immediate-Region fixed effects and state-year fixed effects. Share outcomes are measured in percentage points. Labor-market rows use the full 2010–2024 RAIS panel. Value-added rows use the available IBGE municipal accounts sample through 2021; the regional value-added aggregate is not imputed for 2022–2024. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4: Economic magnitudes of predicted exposure

Outcome	Effect	Baseline mean	Implied change at mean	Baseline median	Implied change at median
Formal employment	+10.15%	2,923 jobs	+297 jobs	162 jobs	+16 jobs
Private establishments	+3.32%	246 estab.	+8 estab.	38 estab.	+1.3 estab.
Real private payroll	+11.00%	R\$ 7.02 mi	+R\$ 0.77 mi	R\$ 0.28 mi	+R\$ 30.7k
Employment share	+0.462 p.p.	8.75 p.p.	+5.3% of mean share	1.45 p.p.	+32.0% of median share
Establishment share	+0.195 p.p.	8.75 p.p.	+2.2% of mean share	2.39 p.p.	+8.1% of median share
Payroll share	+0.480 p.p.	8.75 p.p.	+5.5% of mean share	1.34 p.p.	+35.7% of median share
Regional employment	+11.05%	33,422 jobs	+3,694 jobs	7,665 jobs	+847 jobs
Regional establishments	+4.31%	2,808 estab.	+121 estab.	951 estab.	+41 estab.
Regional payroll	+10.21%	R\$ 80.27 mi	+R\$ 8.20 mi	R\$ 14.18 mi	+R\$ 1.45 mi

Notes: Log coefficients are converted as $100[\exp(\beta) - 1]$. Implied changes multiply the percent effect by baseline 2010 means and medians. Share outcomes are measured in percentage points; relative share changes divide the coefficient by the corresponding baseline share. Monetary values are real. Estab. denotes establishments.

Table 5: Regional creation and value-added intensity

Outcome	Coef.	SE	Observations
<i>Panel A. Regional aggregates</i>			
Log regional employment	0.105***	(0.016)	2,490
Log regional establishments	0.042***	(0.009)	2,490
Log regional payroll	0.097***	(0.020)	2,490
Log regional value added	-0.006	(0.062)	1,992
<i>Panel B. Regional intensity</i>			
Log employment per establishment	0.063***	(0.015)	2,490
Log payroll per worker	-0.008	(0.009)	2,490
Log value added per worker	-0.111*	(0.060)	1,992
Log value added per establishment	-0.048	(0.063)	1,992

Notes: Each row reports the coefficient on standardized region-level predicted exposure. All specifications include Immediate-Region fixed effects and state-year fixed effects. Value-added rows use the available IBGE municipal accounts sample through 2021. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

lishment rise, indicating that local gains are not fully offset elsewhere in the functional region. Payroll per worker, regional value added, value added per worker, and value added per establishment do not rise. The evidence therefore supports a narrower policy interpretation: subsidized credit expands the regional formal employment base and wage bill, but it does not generate measured value-added upgrading.

For policy interpretation, this contrast links the paper to two strands of the place-based literature. Some policies are evaluated as job-creation tools, while others are justified through agglomeration, productivity spillovers, or productive upgrading [Greenstone et al., 2010, Kline and Moretti, 2014,

Lu et al., 2019, Duranton and Venables, 2018]. The results support the first margin more strongly than the second. If the policy objective is to decentralize formal jobs and establishments, the estimates point to a sizeable effect. If the objective is regional productivity growth, credit appears insufficient by itself and may require complementary inputs such as skills, technology adoption, supplier upgrading, or infrastructure.

5.5. Regional hierarchy gradient

Regional hierarchy determines where the formal-scale response is absorbed, since the same regional increase can

reinforce cores or expand lower tiers of the system. A binary core/non-core comparison is informative, but it can hide whether the policy mainly helps large non-core municipalities or reaches lower tiers of the regional system. Table 6 therefore estimates a hierarchy gradient based on each municipality's initial employment share within the Immediate Region.

Moving across the initial regional hierarchy, the coefficients form a clear gradient. Regional cores do not gain employment, and their establishment share falls. High-share non-core municipalities gain regional shares but show limited level effects. Middle-share non-core municipalities expand employment and establishments substantially. Low-share non-core municipalities experience the largest percentage gains in employment and establishments, while also gaining regional shares. The pattern is not simply “outside the core”; effects strengthen among municipalities lower in the initial regional hierarchy.

Agglomeration-based reasoning makes this gradient especially informative because regional cores have thicker labor markets, denser supplier networks, stronger services, and better access to demand. These features make cores natural candidates to absorb subsidized capital [Duranton and Puga, 2004, Combes et al., 2012, Gaubert, 2018]. The estimates point elsewhere. Credit appears to relax constraints at lower tiers of the functional region, allowing less central municipalities to convert subsidized capital into formal jobs and establishments. Because regional shares also rise, the result is not only a small-base log effect. Place-based credit decentralizes formal scale and does not reinforce the existing regional hierarchy.

Geographic evidence supports the interpretation without replacing the regressions because exposure outside cores is widespread enough to rule out a single-city reading of the hierarchy result. In policy terms, the FNE operates as a formal-scale instrument that shifts part of the regional employment base toward less central municipalities.

Table 6: Regional hierarchy gradient by initial regional employment share

Group	Log employment	Employment share	Log establishments	Establishment share
Core/pole	0.019 (0.025)	-0.065 (0.238)	-0.019* (0.010)	-0.648*** (0.146)
Non-core high initial share	0.026 (0.018)	0.540*** (0.128)	-0.001 (0.007)	0.175*** (0.063)
Non-core middle initial share	0.098*** (0.017)	0.519*** (0.079)	0.034*** (0.007)	0.281*** (0.042)
Non-core low initial share	0.173*** (0.020)	0.374*** (0.063)	0.069*** (0.010)	0.203*** (0.034)

Notes: Each cell reports the coefficient on standardized exposure interacted with the group shown in the row. All specifications include municipality fixed effects and Immediate-Region-by-year fixed effects. Regional cores and poles are identified using REGIC and regional hierarchy information. Non-core groups are defined by initial municipal employment share within the Immediate Region. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A Regional cores and poles**B** Exposure among non-core municipalities**Figure 2:** Regional hierarchy and exposure outside regional cores

Notes: Panel A identifies municipalities classified as regional cores or poles. Panel B maps the distribution of predicted exposure among non-core municipalities, while keeping regional cores visible for reference. Darker shades indicate higher exposure among non-core municipalities.

6. Robustness and Shift-Share Diagnostics

Three groups of checks protect the interpretation, with each organized around a specific threat to the design. One group varies exposure construction and tests whether the FNE measure is merely capturing external sectoral employment cycles. A second group addresses shift-share inference and dominant shocks. A third group protects the spatial-incidence estimates from denominator choices, spatial partitions, and large cities. Timing diagnostics are reported in the appendix because exposure is continuous and persistent, not a sharp adoption event.

6.1. Shift-share construction and sector diagnostics

Alternative exposure checks in Table 7 keep the focus on the two cleanest outcomes: log employment and employment share. Panel A varies the treatment definition and adds a control for predicted exposure to external sectoral employment growth. Panel B rebuilds treatment after excluding one broad sector at a time.

Several checks target treatment construction and sectoral confounding before turning to inference and spatial robustness. The estimates do not depend on the non-infrastructure definition, on whether the shock leaves out the state or municipality, on establishment-share exposure, or on the tails of the exposure distribution. External sectoral employment

Table 7: Alternative exposure construction and sector diagnostics

Specification	Log employment	Employment share
<i>Panel A. Alternative exposure construction</i>		
Baseline non-infrastructure	0.097*** (0.015)	0.462*** (0.084)
Total FNE, leave-state-out	0.065*** (0.013)	0.305*** (0.063)
Total FNE, leave-municipality-out	0.065*** (0.013)	0.322*** (0.064)
Establishment-share exposure	0.060*** (0.021)	0.189*** (0.057)
Winsorized baseline exposure	0.108*** (0.016)	0.523*** (0.093)
Baseline with sector-cycle control	0.096*** (0.015)	0.463*** (0.084)
<i>Panel B. Leave-one-sector-out exposure</i>		
Exclude agriculture	0.111*** (0.013)	0.439*** (0.096)
Exclude commerce	0.087*** (0.016)	0.421*** (0.088)
Exclude industry	0.065*** (0.018)	0.390*** (0.086)
Exclude services	0.087*** (0.016)	0.435*** (0.081)
Exclude infrastructure/construction	0.097*** (0.015)	0.462*** (0.084)

Notes: Each entry reports the coefficient on standardized predicted exposure under the specification shown in the row. The sector-cycle specification controls for predicted exposure to external sectoral employment growth. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

cycles also do not explain the result: once exposure to sectoral employment growth is included, the FNE coefficient barely moves. The leave-one-sector-out rows add a second check, since employment and employment-share effects remain positive and precise even when agriculture or industry is removed from the exposure construction.

6.2. Shift-share inference and dominant shocks

The next diagnostic step turns to inference and dominant-shock leverage in Table 8. Panel A compares Immediate-Region clustered standard errors with AKM and AKM0 shift-share standard errors, while Panel B summarizes Rotemberg-style shock-cell weights.

Shift-share inference preserves the main formal-activity interpretation across the outcomes that carry the spatial-incidence result. Employment and employment-share estimates remain precise, and establishment outcomes remain robust under AKM and marginal under AKM0. Rotemberg-style diagnostics show that no single sector-year-state shock-cell dominates the design: the largest absolute weight is 2.65 percent and the top ten account for 19.3 percent. Absolute weights are concentrated in agriculture and industry, yet Table 7 shows that excluding either sector does not overturn the findings. This combination is the relevant defense for a shift-share design: the treatment has identifiable sources of leverage without being a single-shock result.

6.3. Spatial units, denominators, and dominant municipalities

Spatial robustness checks in Table 9 address the possibility that shares move mechanically or that large cities drive the estimates. The first two rows use alternative share constructions, while the remaining rows change fixed effects or restrict the sample.

Across these specifications, the spatial pattern survives because neither denominator changes nor sample restrictions alter the employment and share results. Log share and own-to-rest-of-region measures show that the share result is not a denominator artifact, while broader spatial fixed effects preserve the local employment result. Excluding capitals, regional poles, core municipalities, and the largest initial-employment municipalities leaves the pattern intact, which aligns with the hierarchy-gradient evidence in Section 5.5.

Table 8: Shift-share inference and dominant-shock diagnostics

	Coef.	RGI SE	AKM SE	AKM0 SE
<i>Panel A. Shift-share inference</i>				
Log employment	0.097***	(0.015)	(0.034)***	(0.035)***
Employment share	0.462***	(0.084)	(0.158)***	(0.174)***
Log establishments	0.033***	(0.006)	(0.014)**	(0.017)*
Establishment share	0.195***	(0.040)	(0.076)**	(0.101)*
<i>Panel B. Rotemberg-style diagnostics</i>				
Shock-cells		825		
Largest absolute shock-cell weight		2.65%		
Top 5 absolute shock-cell weights		11.4%		
Top 10 absolute shock-cell weights		19.3%		
Agriculture absolute-weight share		61.3%		
Industry absolute-weight share		22.9%		
Commerce absolute-weight share		11.8%		
Services absolute-weight share		3.9%		

Notes: Panel A reports the baseline coefficient and alternative standard errors for the formal outcomes that carry the main spatial-incidence interpretation. AKM and AKM0 standard errors follow the shift-share inference approach of Adão et al. [2019]. Stars in AKM and AKM0 columns refer to p-values computed using the corresponding standard error. Panel B reports diagnostics based on absolute Rotemberg-style shock-cell weights. Significance levels: *** p<0.01, ** p<0.05, * p<0.10.

Table 9: Robustness to spatial units, denominators, and dominant municipalities

Specification	Log employment	Employment share
Log regional share outcome	0.101*** (0.015)	—
Log own activity relative to rest of region	0.105*** (0.016)	—
Municipality FE + state-year FE	0.094*** (0.015)	0.357*** (0.065)
Municipality FE + Immediate-Region-year FE	0.096*** (0.016)	0.398*** (0.089)
Exclude state capitals	0.096*** (0.015)	0.459*** (0.084)
Exclude regional poles	0.088*** (0.017)	0.297*** (0.094)
Exclude core or pole municipalities	0.091*** (0.017)	0.332*** (0.098)
Exclude top 1% by initial employment	0.096*** (0.015)	0.458*** (0.085)

Notes: The first two rows use alternative outcome constructions to address denominator concerns. Remaining rows change the fixed effects or estimation sample. Standard errors clustered at the relevant regional level are shown in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.10.

7. Conclusion

Place-based credit affects regional economies through several spatial margins, and those margins determine the meaning of a local gain. A municipality receiving stronger exposure may grow because the region creates new formal activity, because activity moves across nearby places, or because the policy changes the municipality's position within the regional hierarchy. Using the FNE as a large place-based credit program, this paper separates those margins in a municipality-year panel organized around functional Immediate Regions.

Once the four spatial margins are considered together, the central result is clear. Predicted exposure to subsidized credit raises formal employment, establishments, and payroll in exposed municipalities; the rest of the Immediate Region does not contract; regional formal aggregates grow; and exposed municipalities gain shares of employment, establishments,

and payroll. Magnitudes are economically meaningful, corresponding to about 10 percent higher municipal employment and 11 percent higher regional employment for a one-standard-deviation increase in exposure.

The hierarchy evidence then shows where this new scale appears within the functional regional economy. Effects are small in regional cores and strongest among initially less central or smaller non-core municipalities. Place-based credit therefore decentralizes formal scale inside functional regional economies and does not simply reinforce established centers. Value-added outcomes discipline the interpretation: regional value added and value-added intensity do not rise, so the evidence supports formal-scale creation and decentralization, not regional productivity upgrading.

Consequently, the regional-science interpretation is specific and disciplined by the contrast between formal scale and value-added intensity. Territorial finance can expand formal employment opportunities outside regional cores and

increase the regional wage base, but credit alone need not raise value-added intensity. If the objective is to decentralize formal jobs, place-based credit can be a meaningful instrument; if the objective is regional productivity growth, credit may require complementary policies that raise technology adoption, worker skills, supplier quality, and firm capabilities. Evaluations of territorial finance should therefore ask not only whether treated municipalities grow, but how functional regional economies absorb that growth and which places inside the hierarchy capture it.

A. Exposure Construction and First-Stage Diagnostics

Appendix A documents exposure construction and first-stage relevance, showing that predicted exposure is empirically connected to observed FNE credit and that its variation is not concentrated in a single sector-year cell.

Table 10: Municipal first-stage variants

Outcome	Coef.	SE	Observations
Log FNE non-infrastructure	0.077**	(0.032)	28,470
Log FNE total	0.068**	(0.032)	28,470
FNE non-infrastructure share in region	0.295*	(0.164)	28,470
FNE total share in region	0.214*	(0.123)	28,470

Notes: Each row reports the coefficient on standardized predicted exposure in a municipality-level first-stage specification with municipality fixed effects and Immediate-Region-by-year fixed effects. Share outcomes are measured in percentage points. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 11: Absolute contribution to exposure by sector

Sector	Share of absolute contribution
Agriculture	39.2%
Commerce	30.2%
Services	15.5%
Industry	15.1%
Infrastructure/construction	0.0%

Notes: Contributions are computed from the absolute sectoral components of the baseline predicted exposure before standardization.

Table 12: Largest sector-year exposure contributions

Sector-year	Year	Share of absolute contribution
Agriculture	2024	4.5%
Agriculture	2023	4.4%
Agriculture	2022	3.8%
Commerce	2024	3.4%
Commerce	2020	3.0%
Commerce	2023	2.8%
Commerce	2019	2.7%
Agriculture	2021	2.7%
Commerce	2018	2.4%
Agriculture	2019	2.4%

Notes: The table lists the ten largest sector-year contributions to the absolute exposure component. No sector-year accounts for more than 4.5% of the absolute contribution.

B. Shift-Share Robustness and Inference

Appendix B expands the shift-share diagnostics reported in the main text, showing that formal labor-market findings survive alternative treatment construction, sector-cycle controls, shift-share inference, and dominant-shock diagnostics.

Table 13: Sector-cycle controls for formal outcomes

Outcome and term	Coef.	SE
Log employment: FNE exposure	0.096***	(0.015)
Log employment: sector-cycle exposure	-0.004	(0.013)
Employment share: FNE exposure	0.463***	(0.084)
Employment share: sector-cycle exposure	0.024	(0.048)
Log establishments: FNE exposure	0.032***	(0.006)
Log establishments: sector-cycle exposure	-0.006	(0.005)
Establishment share: FNE exposure	0.197***	(0.040)
Establishment share: sector-cycle exposure	0.044**	(0.021)

Notes: Each specification includes both FNE predicted exposure and predicted exposure to external sectoral employment growth. All models include municipality fixed effects and Immediate-Region-by-year fixed effects. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 14: Leave-one-sector-out exposure checks

Excluded sector	Log employment	Employment share
Agriculture	0.111*** (0.013)	0.439*** (0.096)
Commerce	0.087*** (0.016)	0.421*** (0.088)
Industry	0.065*** (0.018)	0.390*** (0.086)
Services	0.087*** (0.016)	0.435*** (0.081)
Infrastructure/construction	0.097*** (0.015)	0.462*** (0.084)

Notes: Each row rebuilds exposure after excluding the sector shown. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 15: Full AKM and AKM0 shift-share inference

Outcome	Coef.	RGI SE	AKM SE	AKM0 SE
Log employment	0.097***	(0.015)	(0.034)***	(0.035)***
Employment share	0.462***	(0.084)	(0.158)***	(0.174)***
Log establishments	0.033***	(0.006)	(0.014)**	(0.017)*
Establishment share	0.195***	(0.040)	(0.076)**	(0.101)*
Log payroll	0.104***	(0.023)	(0.047)**	(0.051)**
Payroll share	0.480***	(0.102)	(0.190)**	(0.194)**
Log value added	0.054***	(0.017)	(0.047)	(0.050)
Value-added share	0.394***	(0.146)	(0.425)	(0.441)

Notes: RGI SE denotes standard errors clustered at the Immediate-Region level. AKM and AKM0 standard errors follow Adão et al. [2019]. Stars in AKM and AKM0 columns are based on the corresponding p-values. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 16: Rotemberg-style exposure-weight diagnostics

Diagnostic	Value
Number of shock-cells	825
Largest absolute shock-cell weight	2.65%
Top 5 absolute shock-cell weights	11.4%
Top 10 absolute shock-cell weights	19.3%
Sum of negative weights	-1.17
Sum of absolute weights	3.34
Agriculture absolute-weight share	61.3%
Industry absolute-weight share	22.9%
Commerce absolute-weight share	11.8%
Services absolute-weight share	3.9%
Infrastructure/construction absolute-weight share	0.0%

Notes: Shock-cells are sector-by-state-leave-out-by-year cells. Values are reported for the employment and employment-share exposure design; weights are the same across the formal outcomes because the same residualized exposure is used.

Table 17: Largest Rotemberg-style shock-cells

Shock-cell	Sector	Year	Absolute weight share
Agriculture-BA-2023	Agriculture	2023	2.65%
Agriculture-BA-2024	Agriculture	2024	2.55%
Agriculture-MA-2024	Agriculture	2024	2.23%
Agriculture-MA-2023	Agriculture	2023	2.09%
Agriculture-PB-2023	Agriculture	2023	1.91%
Agriculture-PB-2024	Agriculture	2024	1.86%
Agriculture-BA-2022	Agriculture	2022	1.82%
Agriculture-MA-2022	Agriculture	2022	1.49%
Agriculture-PB-2022	Agriculture	2022	1.40%
Agriculture-RN-2023	Agriculture	2023	1.36%

Notes: Shock-cells are sector-by-state-leave-out-by-year cells. The table lists the ten largest absolute weights for the employment exposure design.

C. Spatial Accounting and Spatial Robustness

Appendix C reports disaggregated local and share estimates together with additional spatial robustness checks. These results protect the within-region share estimates from denominator choices, alternative spatial partitions, and concentration in a small number of dominant municipalities.

Table 18: Disaggregated local effects of predicted exposure

	Log employment	Log establishments	Log payroll	Log value added
Predicted exposure	0.097*** (0.015)	0.033*** (0.006)	0.104*** (0.023)	0.054*** (0.017)
Observations	28,470	28,470	28,470	22,772
Municipality FE	Yes	Yes	Yes	Yes
Immediate region $\times$ year FE	Yes	Yes	Yes	Yes
Clusters	166	166	166	166

Notes: Each column reports the coefficient on standardized municipal predicted exposure. Standard errors clustered at the Immediate-Region level are shown in parentheses. Value-added outcomes use the available IBGE municipal accounts sample through 2021. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 19: Disaggregated within-region share effects

	Employment share	Establishment share	Payroll share	Value-added share
Predicted exposure	0.462*** (0.084)	0.195*** (0.040)	0.480*** (0.102)	0.394*** (0.146)
Observations	28,470	28,470	28,470	22,776
Municipality FE	Yes	Yes	Yes	Yes
Immediate region $\times$ year FE	Yes	Yes	Yes	Yes
Clusters	166	166	166	166

Notes: Dependent variables are municipal shares of the corresponding Immediate-Region aggregate, measured in percentage points. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 20: Spatial accounting main estimates reproduced for comparison

Outcome	Exposed municipality	Rest of region	Entire region	Within-region share
Employment	0.097*** (0.015)	0.013** (0.006)	0.105*** (0.016)	0.462*** (0.084)
Establishments	0.033*** (0.006)	0.008** (0.004)	0.042*** (0.009)	0.195*** (0.040)
Payroll	0.104*** (0.023)	0.012 (0.008)	0.097*** (0.020)	0.480*** (0.102)
Value added	0.054*** (0.017)	0.024** (0.010)	-0.006 (0.062)	0.394*** (0.146)

Notes: This table reproduces the main spatial-accounting estimates to facilitate comparison with the additional spatial checks below. Labor-market rows use the full 2010–2024 RAIS panel. Value-added rows use the available IBGE municipal accounts sample through 2021; the regional value-added aggregate is not imputed for 2022–2024.

Table 21: Robustness to spatial units, denominators, and dominant municipalities

Specification	Log employment	Employment share
Log regional share outcome	0.101*** (0.015)	—
Log own activity relative to rest of region	0.105*** (0.016)	—
Municipality FE + state-year FE	0.094*** (0.015)	0.357*** (0.065)
Municipality FE + Immediate-Region-year FE	0.096*** (0.016)	0.398*** (0.089)
Exclude state capitals	0.096*** (0.015)	0.459*** (0.084)
Exclude regional poles	0.088*** (0.017)	0.297*** (0.094)
Exclude core or pole municipalities	0.091*** (0.017)	0.332*** (0.098)
Exclude top 1% by initial employment	0.096*** (0.015)	0.458*** (0.085)

Notes: The first two rows use alternative outcome constructions to address denominator concerns. Remaining rows change the fixed effects or estimation sample. Standard errors clustered at the relevant regional level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 22: Regional concentration outcomes

Outcome	Coef.	SE
Employment Herfindahl index	0.009	(0.006)
Establishment Herfindahl index	0.000	(0.002)
Pole employment share	0.437	(0.597)
Pole establishment share	0.081	(0.256)

Notes: Outcomes are measured at the Immediate-Region-year level. Specifications include Immediate-Region fixed effects and state-year fixed effects. Standard errors clustered at the Immediate-Region level are shown in parentheses.

D. Mechanisms, Scale, Value-Added Intensity, and Economic Magnitudes

Appendix D reports the scale and intensity outcomes behind the interpretation in Section 5.4 together with the extended economic-magnitude calculations. The tables show that exposure expands formal scale more consistently than it raises payroll or value-added intensity.

Table 23: Municipal scale and intensity outcomes

Outcome	Coef.	SE
Log employment	0.176***	(0.021)
Log establishments	0.050***	(0.009)
Log employment per establishment	0.126***	(0.018)
Log payroll per worker	0.011	(0.018)
Log value added per worker	-0.122***	(0.024)
Log value added per establishment	0.004	(0.019)

Notes: All specifications use the common municipal value-added sample and include municipality fixed effects and Immediate-Region-by-year fixed effects. Employment and establishment coefficients in this table should be read as within-sample scale diagnostics, not as the paper's main magnitudes. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 24: Regional scale and intensity outcomes

Outcome	Coef.	SE
Log regional employment	0.105***	(0.016)
Log regional establishments	0.042***	(0.009)
Log employment per establishment	0.063***	(0.015)
Log payroll per worker	-0.008	(0.009)
Log value added per worker	-0.111*	(0.060)
Log value added per establishment	-0.048	(0.063)

Notes: Specifications use Immediate-Region-year outcomes and include Immediate-Region fixed effects and state-year fixed effects. Value-added rows use the available IBGE municipal accounts sample through 2021. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 25: Extended economic magnitudes

Outcome	Effect	Baseline mean	Implied change at mean	Baseline median	Implied change at median
Formal employment	+10.15%	2,923 jobs	+297 jobs	162 jobs	+16 jobs
Private establishments	+3.32%	246 estab.	+8 estab.	38 estab.	+1.3 estab.
Real private payroll	+11.00%	R\$ 7.02 mi	+R\$ 0.77 mi	R\$ 0.28 mi	+R\$ 30.7k
Employment share	+0.462 p.p.	8.75 p.p.	+5.3% of mean share	1.45 p.p.	+32.0% of median share
Establishment share	+0.195 p.p.	8.75 p.p.	+2.2% of mean share	2.39 p.p.	+8.1% of median share
Payroll share	+0.480 p.p.	8.75 p.p.	+5.5% of mean share	1.34 p.p.	+35.7% of median share
Regional employment	+11.05%	33,422 jobs	+3,694 jobs	7,665 jobs	+847 jobs
Regional establishments	+4.31%	2,808 estab.	+121 estab.	951 estab.	+41 estab.
Regional payroll	+10.21%	R\$ 80.27 mi	+R\$ 8.20 mi	R\$ 14.18 mi	+R\$ 1.45 mi

Notes: Log coefficients are converted as $100[\exp(\beta) - 1]$. Implied changes multiply the percent effect by baseline 2010 means and medians. Share outcomes are measured in percentage points; relative share changes divide the coefficient by the corresponding baseline share. Monetary values are real.

E. Regional Hierarchy Gradients and Exposure Persistence Diagnostics

Appendix E expands the hierarchy evidence and reports timing diagnostics, completing the checks on where and when exposure is absorbed. The gradient by initial size complements the main gradient by initial regional employment share, while the exposure-persistence estimates are not interpreted as a staggered-adoption event study because treatment is continuous and serially correlated.

Table 26: Regional hierarchy gradient by initial employment size

Group	Log employment	Employment share	Log establishments	Establishment share
Core/pole	0.020 (0.025)	-0.072 (0.238)	-0.019* (0.010)	-0.652*** (0.145)
Non-core large initial employment	0.028 (0.018)	0.425*** (0.125)	-0.002 (0.006)	0.103* (0.057)
Non-core medium initial employment	0.088*** (0.017)	0.509*** (0.082)	0.025*** (0.007)	0.236*** (0.048)
Non-core small initial employment	0.203*** (0.020)	0.503*** (0.071)	0.091*** (0.010)	0.342*** (0.039)

Notes: Each cell reports the coefficient on standardized exposure interacted with the group shown in the row. All specifications include municipality fixed effects and Immediate-Region-by-year fixed effects. Non-core groups are defined by initial private employment. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 27: Binary regional hierarchy benchmark

	Log employment	Employment share	Log establishments	Establishment share
Predicted exposure	0.099*** (0.015)	0.476*** (0.084)	0.034*** (0.006)	0.217*** (0.040)
Exposure $\times$ core/pole	-0.075*** (0.016)	-0.547** (0.224)	-0.052*** (0.009)	-0.865*** (0.145)
Implied effect in core/pole	0.023 (0.025)	-0.070 (0.238)	-0.018* (0.010)	-0.648*** (0.146)
Observations	28,470	28,470	28,470	28,470
Municipality FE	Yes	Yes	Yes	Yes
Immediate region $\times$ year FE	Yes	Yes	Yes	Yes
Clusters	166	166	166	166

Notes: The interaction term allows the exposure effect to differ between regional cores/poles and other municipalities. The first row reports the effect for non-core municipalities. The third row reports the implied total effect in cores/poles. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 28: Regional hierarchy: payroll outcomes

	Log payroll	Payroll share
Predicted exposure	0.107*** (0.023)	0.486*** (0.101)
Exposure $\times$ core/pole	-0.081*** (0.019)	-0.246 (0.270)
Implied effect in core/pole	0.025 (0.033)	0.240 (0.292)

Notes: Specifications include municipality fixed effects and Immediate-Region-by-year fixed effects. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 29: Exposure persistence diagnostics for formal outcomes

Outcome	Current exposure	Lead $t + 1$	Lead $t + 2$
Log employment	0.104*** (0.017)	0.033*** (0.010)	0.002 (0.014)
Log establishments	0.031*** (0.007)	0.008* (0.005)	0.007 (0.007)
Log payroll	0.107*** (0.028)	0.032** (0.016)	0.011 (0.021)
Employment share	0.454*** (0.108)	0.142* (0.073)	0.022 (0.073)
Establishment share	0.097*** (0.035)	0.056*** (0.021)	0.076** (0.031)
Payroll share	0.497*** (0.137)	0.139 (0.091)	0.018 (0.089)

Notes: Each row group comes from a specification including contemporaneous exposure and two future values of exposure. Future exposure is included to characterize persistence in the exposure process, not as a pre-treatment falsification test. Because exposure is continuous and serially correlated, these coefficients should not be interpreted as event-study leads. All specifications include municipality fixed effects and Immediate-Region-by-year fixed effects. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 30: Distributed lags for formal outcomes

Outcome	Current exposure	Lag $t - 1$	Lag $t - 2$
Log employment	0.045*** (0.013)	0.017 (0.010)	0.040** (0.015)
Log establishments	0.025*** (0.006)	0.006 (0.005)	0.005 (0.007)
Log payroll	0.066*** (0.018)	0.012 (0.015)	0.033 (0.022)
Employment share	0.267*** (0.070)	0.080 (0.060)	0.081 (0.105)
Establishment share	0.149*** (0.033)	0.067*** (0.022)	-0.027 (0.038)
Payroll share	0.251*** (0.091)	0.102 (0.070)	0.078 (0.137)

Notes: Each row group comes from a specification including contemporaneous exposure and two lags. All specifications include municipality fixed effects and Immediate-Region-by-year fixed effects. Standard errors clustered at the Immediate-Region level are shown in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

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